



NATIONAL POLYTECHNIC INSTITUTE OF
CAMBODIA

FACULTY OF ELECTRONIC

DEPARTMENT OF ELECTRONIC

Assignment

Power Electronic Lab

Testing Power Supply 60W
Model: S-60-12

Professor's Name :

Meas Saran

Student Name :

Phon Phearon

2024 - 2025

Contents

1	Introduction	1
1.1	Power Supply	1
1.2	Purpose	1
2	Power Supply Model S-60-12	1
2.1	Power Supply Model S-60-12 Overview	1
2.2	Power Supply Model S-60-12 Topology	2
2.2.1	Flyback Converter Working Principle	2
2.3	Power Supply Model S-60-12 Schematic	3
2.4	Power Supply Model S-60-12 Main Components	5
3	Testing Power Supply S-60-12	6
3.1	Testing With Load	6

1 Introduction

1.1 Power Supply

A **power supply** is a vital component in any electrical or electronic system, responsible for delivering the necessary electrical energy for devices to function. It converts electrical energy from a source—such as a wall outlet, battery, or generator—into a usable form with the correct voltage, current, and frequency for the load it powers.

Power supplies come in various types, including **linear** and **switch-mode** power supplies, each suited for different applications based on factors like efficiency, size, and noise. They may provide **AC (alternating current)** or **DC (direct current)** output, depending on the needs of the connected device.

In both consumer electronics and industrial applications, the reliability and performance of the power supply directly impact the operation of the entire system. As such, understanding the principles, types, and design of power supplies is fundamental in the fields of electrical engineering and electronics.

1.2 Purpose

The **purpose** of this report is to provide a technical overview and analysis of the S-60-12 power supply module. This document aims to highlight the key features, operating principles, and performance characteristics of the module, making it useful for engineers, technicians, and students involved in electronics and power system applications.

2 Power Supply Model S-60-12

2.1 Power Supply Model S-60-12 Overview

The **S-60-12** is a commonly used switching power supply (also known as an SMPS - Switched-Mode Power Supply). It's often used in electronics, LED strip lighting, CCTV systems, and other low-voltage applications. Here's a breakdown of the S-60-12 model:



Figure 1: *Power Supply S-60-12*

- **Manufacturer Style:** It follows the naming convention used by companies like Mean Well and other generic power supply makers.

- **Switching Power Supply (AC to DC converter).**
- **Cooling** Typically air-cooled (via natural convection; some models have a small fan but S-60-12 usually doesn't)

Specification of Power Supply Model S-60-12

Table 1: Specifications of Model S-60-12 60W Power Supply

Feature	Description
Input Voltage	100–240V AC, 50/60 Hz
Output Voltage	12V DC
Output Current	5 Amps
Power Output	60 Watts ($12V \times 5A$)
Efficiency	Around 80–85%
Adjustable Output	Usually has a small potentiometer to tweak voltage (approx $\pm 10\%$)
Protection Features	Overload, Overvoltage, and Short-Circuit Protection
Enclosure Type	Metal case with ventilation holes (perforated)

2.2 Power Supply Model S-60-12 Topology

The S-60-12 switching power supply typically uses a flyback converter topology, which is very common in low-to-mid power applications (like this 60W model).

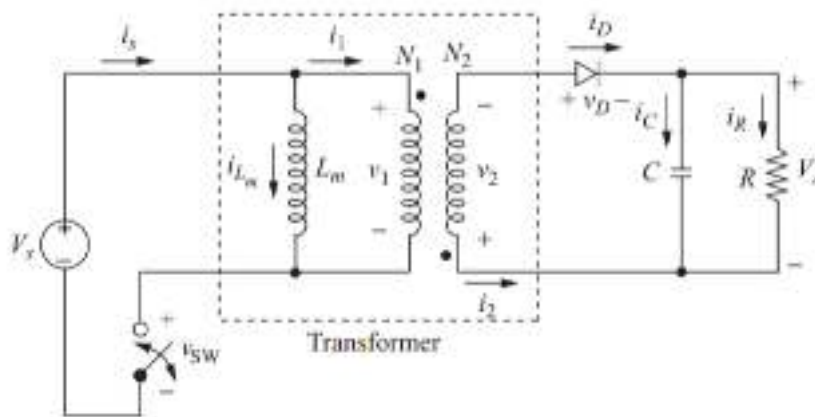


Figure 2: *Flyback Converter Topology*

2.2.1 Flyback Converter Working Principle

1. AC Input Stage:

- Full bridge rectifier + filter capacitor convert AC (100–240V) to high-voltage DC (310V for 220V input).
- This is the DC bus for the switching stage.

2. High-Frequency Switching:

- A PWM controller drives a power MOSFET.
- The MOSFET rapidly switches on and off (typically 30–100 kHz), sending pulses into the flyback transformer.

3. Flyback Transformer:

- Stores energy in its magnetic field when the MOSFET is on.
- When the MOSFET turns off, energy is transferred to the secondary winding and rectified to DC.

4. Output Rectification & Filtering:

- The transformer secondary outputs 12V DC after passing through a fast diode and electrolytic capacitors for filtering.

5. Feedback Control:

- An optocoupler and reference IC (like TL431) provide feedback from the output to the PWM controller.
- This feedback stabilizes the output voltage and maintains regulation under varying loads.

2.3 Power Supply Model S-60-12 Schematic

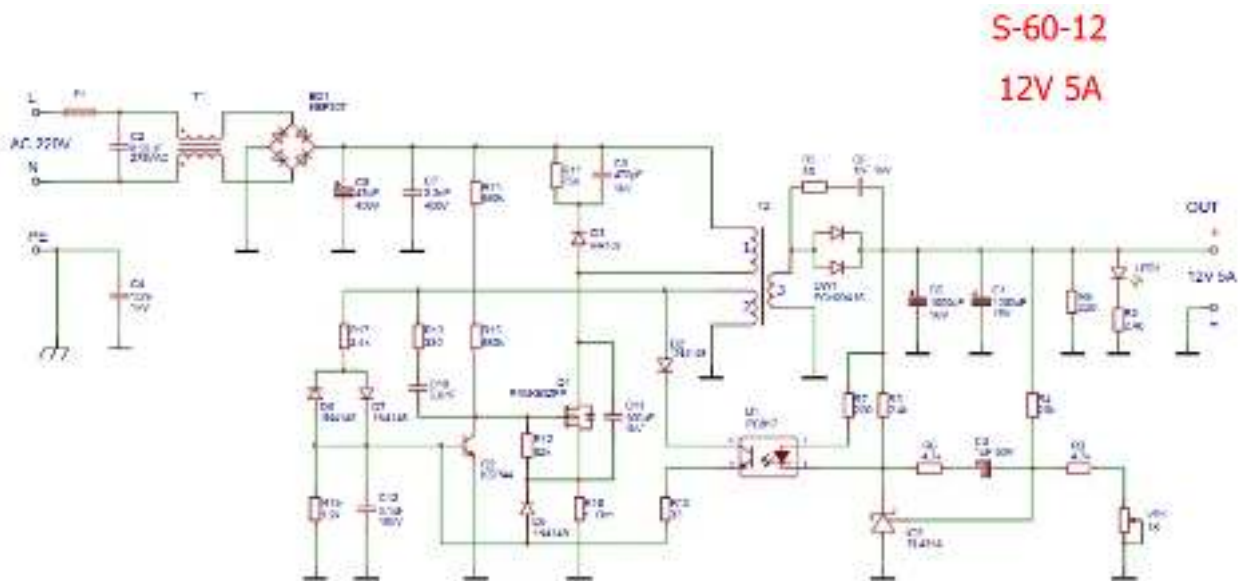


Figure 3: *Power Supply Model S-60-12 Schematic*

The schematic diagram represents a flyback converter-based power supply designed to convert AC 220V input to a regulated 12V DC output capable of delivering 5A .

1. Input Section (AC Input L/N/PE):

- **F1 (Fuse):** Provides overcurrent protection for the primary circuit.
- **C2 (2.2μF/275VAC):** Acts as an EMI filter to suppress high-frequency noise from the mains.
- **T1 (Transformer):** Steps down the AC voltage to a lower level (15-20VAC) for the primary side.

2. Rectification & Filtering (Primary Side):

- **BD1 (KBPC307 Bridge Rectifier):** Converts stepped-down AC voltage from T1 into pulsating DC.
- **C5 (470 μ F/400V) & C7 (2.2 μ F/400V):** Smooth the rectified DC voltage, reducing ripple.
- **R11 (33 Ω) & R14 (820 Ω):** Form a voltage divider to monitor primary-side voltage.
- **D5 (FR107 Diode):** Protects against reverse polarity or transient voltages.

3. Switching Circuit:

- **Q1 (PAN902ZFP MOSFET):** Primary switch controlled by PWM signal from control circuit.
- **T2 (Flyback Transformer):**
 - *Primary Winding:* Carries switched current from Q1.
 - *Secondary Winding:* Transfers energy to secondary side when Q1 turns off.
 - *Auxiliary Winding (Pin 3):* Supplies power to control circuit after startup.

4. Secondary Side Rectification & Filtering:

- **D1 (FR207 Diode):** Rectifies induced AC voltage from T2's secondary winding.
- **C5 (1000 μ F/16V) & C1 (1000 μ F/16V):** Filter rectified DC to ensure stable 12V output.
- **LED1 (220 Ω Resistor):** Indicates power presence at the output.

5. Feedback Loop & Regulation:

- **U1 (PC817 Optocoupler):** Isolates primary/secondary sides while transmitting error signals.
- **IC2 (TL431 Shunt Regulator):** Generates precise 2.5V reference voltage.
- **VR1 (1k Ω Potentiometer):** Adjusts output voltage by altering feedback voltage.
- **R3 (4.7k Ω) & R4 (4.7k Ω):** Set voltage divider ratio for feedback.
- **Control Mechanism:**
 - TL431 compares output voltage (via R3/R4) to 2.5V reference
 - Modulates optocoupler LED current to adjust Q1's duty cycle

6. Startup & Biasing:

- **R16 (1.2k Ω) & C12 (1 μ F/100V):** Provide initial gate voltage to Q1.
- **D7 (IN4148 Diode) & C9 (100nF):** Supply power to control circuit after auxiliary winding activation.

7. Protection Features:

- **DW1 (PCH20A15 TVS):** Clamps overvoltages on secondary side.
- **D6 (IN4148 Diode):** Dissipates stored energy in T2's primary winding when Q1 turns off.

2.4 Power Supply Model S-60-12 Main Components

Key Components & Roles

Table 2: Key Components and Roles

Component	Role
T2 (Flyback Transformer)	Energy transfer between primary/secondary sides
Q1 (MOSFET)	High-frequency switching (50–100kHz)
PC817 (Optocoupler)	Galvanic isolation for feedback
TL431	Output voltage regulation
F1 (Fuse)	Overcurrent protection

1. Controller

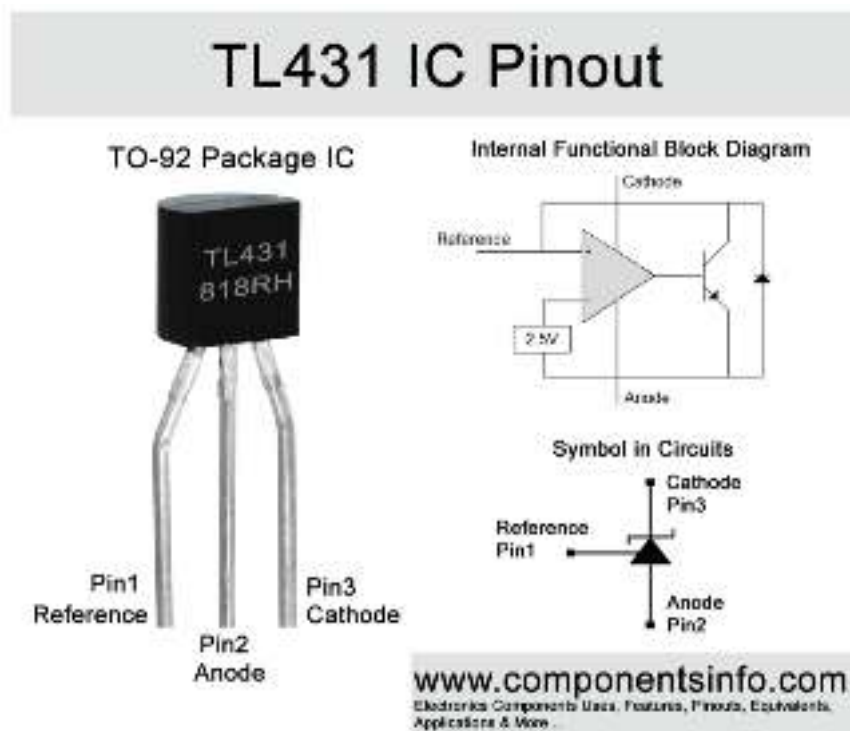


Figure 4: *TL431 IC*

2. Opto Coupler

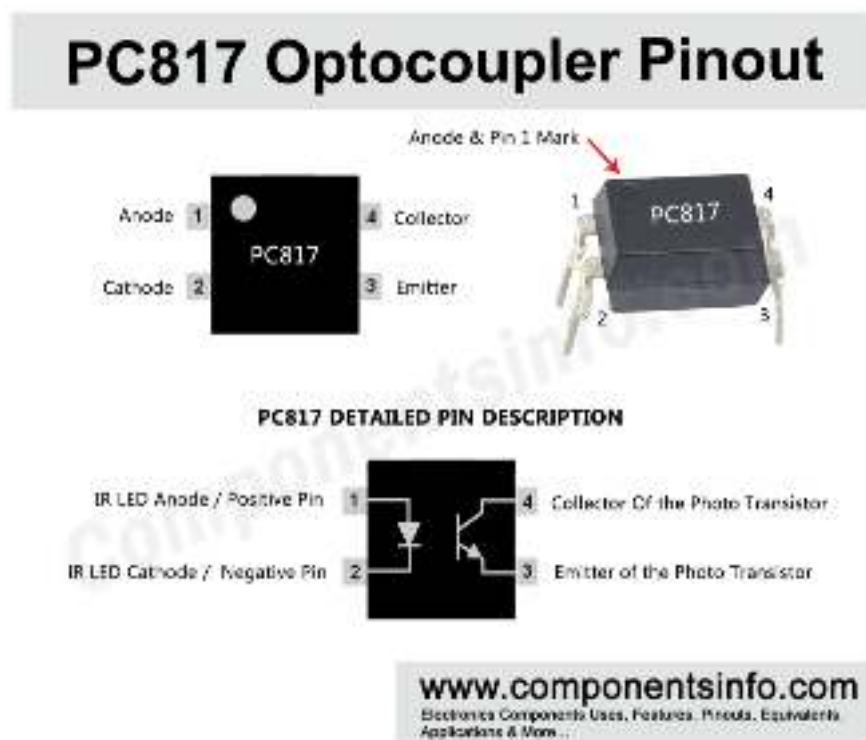


Figure 5: *PC817*

3 Testing Power Supply S-60-12

3.1 Testing With Load



Figure 6: *Testing Setup*

- 10W Testing



Figure 7: *Graph Duty Cycle and Inductor Current [10W]*

- 20W Testing



Figure 8: *Graph Duty Cycle and Inductor Current [20W]*

- 30W Testing

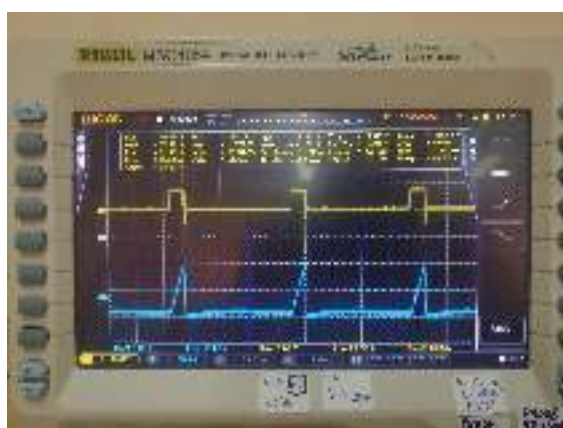


Figure 9: *Graph Duty Cycle and Inductor Current [30W]*

- 40W Testing



Figure 10: *Graph Duty Cycle and Inductor Current [40W]*

- 50W Testing



Figure 11: *Graph Duty Cycle and Inductor Current [50W]*

- 60W Testing



Figure 12: *Graph Duty Cycle and Inductor Current [60W]*